MAHMOUD MOHAMED MONTASER HELMY

Artificial Intelligence Engineer

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience designing, developing, and deploying Deep Learning, Computer Vision, and Generative AI solutions. Proven ability to build high-performance, real-time AI applications with a strong focus on accuracy, scalability, and production readiness, with experience developing RESTful APIs and integrating AI models into modern web and backend systems. Skilled in managing end-to-end AI project lifecycles, from research and data preparation to model development, evaluation, API integration, and deployment, Strong focus on delivering practical, performance-driven AI solutions that translate machine learning research into reliable real-world applications..

TECHNICAL SKILLS

Programming: Python, C++, SQL, Bash, MATLAB
AI/ML: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face
NLP & LLMs: Transformers, RAG, LangChain, LlamaIndex, LoRA/QLoRA, Prompt Engineering, FAISS, ChromaDB
Agentic AI: LangGraph, AutoGen, CrewAI, Function Calling, Multi-Agent Systems
Computer Vision: OpenCV, YOLO, CNNs, Object Detection, Segmentation, OCR, Facial Recognition
Generative AI: GPT, LLaMA, Mistral, Stable Diffusion, GANs, Quantization, vLLM
MLOps & Deployment: Docker, FastAPI, Flask, MLflow, ONNX, Azure ML, REST APIs
Data , Cloud & Tools: NumPy, Pandas, Apache Spark, AWS, Azure, Git, Linux, Jupyter Notebook

SOFT SKILLS

Professional Skills: Communication, Teamwork, Problem Solving

PROFESSIONAL EXPERIENCE

Artificial Intelligence Engineer - Freelance **Jun 2023 – Present**

- Developed and optimized **ML/DL and Computer Vision models** for real-time AI applications, including facial recognition, object detection, and medical image classification.
- Built and deployed **REST APIs** for scalable AI inference across different platforms and technologies.
- Conducted model evaluation, architecture comparison, and **hyperparameter tuning**
- Managed **end-to-end AI projects**, from data preprocessing and augmentation to deployment and delivery.

PROJECTS

Alzheimer's Disease Detection | Graduation Project

- Led the AI development of a medical-imaging project focused on Alzheimer's disease detection.
- Implemented (DCGAN) to generate synthetic medical images for dataset augmentation.
- Built a (CNN) model for medical-image classification, achieving 98% accuracy.

Real-Time Emotion Recognition

- Developed a real-time computer-vision application that detects and analyzes facial expressions using deep learning.
- Integrated a CNN model with a Flask web platform, achieving 98% accuracy in emotion classification.

AI-Powered Attendance System

- Developed a real-time attendance application using facial recognition and (LBPH) model.
- Integrated the recognition component with a .NET backend and web-based frontend for attendance management.

EDUCATION

Higher Technological Institute - Sharqia, Egypt **Oct 2021 – Jan 2025**

B.Sc. in Computer Science | GPA: 3.52

CERTIFICATIONS

- Practical Machine Learning for Data Scientists - Coursat.ai
- AI and Machine Learning - CLS
- Internet of Things (IoT) - ITI

LANGUAGES

Arabic - Native | English - Native